Isolation Forest algorithm for anomaly detection

Arpit Narwal
Mar 1, 2022
5 min read

Anomaly detection is identifying something that could not be termed as "normal", the definition of "normal" depends on the phenomenon that is being observed. Isolation Forest algorithm is a tree based algorithm that uses an ensemble approach to detect anomalies. Unlike other Machine Learning algorithms Isolation Forest algorithm does not explicitly profile the characteristics of anomalies, keeping it independent of data distributions making this approach novel.

Characteristics of anomalies

Since anomalies deviate from normal, they are few in numbers (minority) and to have attributes values that are very different from the normal. The paper nicely puts it as **few and different**. These characteristics of anomalies are the foundation for the Isolation Forest algorithm and form the guiding principle of the Isolation Forest algorithm.

A I Conference
Artificial Intelligence, ML, Data Science, Robotics, Quantum, Blockchain

Trending AI Articles:

1. Basics of Neural Network
2. Are you using the term 'AI' incorrectly?
3. How Can We Improve the Quality of Our Data?
4. Artificial Intelligence Conference

The issue with the usual approach

The usual methods are not optimized to detect anomalies, instead they are optimized to profile normal instances. Most of the usual methods can detect anomalies as a by-product of how they define normal but they can be inaccurate. Many of these methods are computationally complex and hence not ideal for anomaly detection.

Isolation Forest algorithm addresses both of the above concerns and provides an efficient and accurate way to detect anomalies.

The algorithm

Now we take a go through the algorithm, and dissect it stage by stage and in the process understand the model behind it. Forest, your next bite, it's going to be a hungry ride.

The core principle

The core of the algorithm is to "isolate" anomalies by creating decision trees over random attributes. The random partitioning produces noticeable shorter paths for anomalies.

- fewer instances (of anomalies) result in smaller partitions
- distinguishable attribute values are more likely to be separated in early partitioning.

Hence, when a forest of random trees collectively produces shorter path lengths for some particular points, then they are highly likely to be anomalies.

The diagram above shows the number of splits required to isolate a normal point and an anomaly. Before, representing overhead views, the hypothesis is random on a random attribute and in the process building a decision tree. The number of splits (decisions) required for a which the anomaly is isolated and will be used to generate the anomaly score.

Isolation Forest algorithm builds a forest of such random trees and averages the anomaly score of the given point. Subsampling is used to create these trees. The tree is created until the isolation of the average level of which the point is isolated. The top c_{ave} performed on the basis of the score, are labeled as anomalies.

Construction of decision tree

The decision tree is created by randomly selecting the sub-sample points/instances over a split value of a randomly selected attribute each. Run the iteration recursively, corresponding overhead views, until the sub-sample value point and the attribute graph, and the process is continued until the tree reaches its height limit. The height limit is set to the average of the instances the minimum and maximum values of the selected attribute.

These are two types of nodes in the decision tree

Internal Node

Internal nodes have one left and one/ one/ right. Internal nodes are used for partitions to non-leaf nodes. An internal node is always a parent to two nodes and has one splitting the entire decision tree a proper binary tree.

External Node

External nodes are leaf nodes that could not be split further and ended at the end of the tree. The path length of an external node is the number of edges which is used to calculate the anomaly score.

The image above shows how sub-sampling actually makes a clear separation between normal points and anomalies. In the original dataset, the anomaly is very close to the normal points, but after sub-sampling the anomaly and normal points look very far and distinguishable.

Why sub-sampling helps

The Isolation Forest algorithm works well when the trees are created, not from the entire dataset, but from a sub-sample of the dataset. This is a different from almost all other techniques where they derive on data and then use the entire dataset to profile normal points. Sub-sampling helps in two ways, it reduces the swamping and masking effects with the anomalies. Swamping is being a little closer to the anomalies, masking is when anomalies are close to the target point and so on (anomalies) seem that there could not be any anomalies.

```
# sub-sample function, returns sample, sub-sample
def sub_sample(dataset, sub_sample_size):
    """Takes dataset and sub-sample size as input and returns a
    sample, sub-sample
    Keyw arg: dataset: dataset to be sub-sampled
              sub_sample_size: size of the sub-sample to be taken
    Ret: sample: the sample of the data that was not selected
         sub_sample: the sample of the data that was selected
    """
    # sub-sample
    sub_sample = dataset.sample(sub_sample_size)
    # sample
    sample = dataset.drop(sub_sample.index)
    return sample, sub_sample

# sub-sample size
sub_sample_size = 256(DEFAULT)
# to set of trees to 100 (as has an of 0.9999 score)
# and average path length of normal points to 10
# and average path length of anomalous points to 2
# and average path length of anomalous points to 2
# and average path length of anomalous points to 2
```